



INTELLIGENT PESTICIDE SPRINKLING SYSTEM

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INTRODUCTION

- Farmers often spray pesticides uniformly without knowing the infection level.
- This leads to wastage, higher costs, and environmental damage.
- A smart system using AI and IoT can solve this problem.



PROBLEM STATEMENT

- Farmers cannot accurately detect infection levels.
- Excess pesticide spraying increases cost and pollution
- No automated monitoring and control system

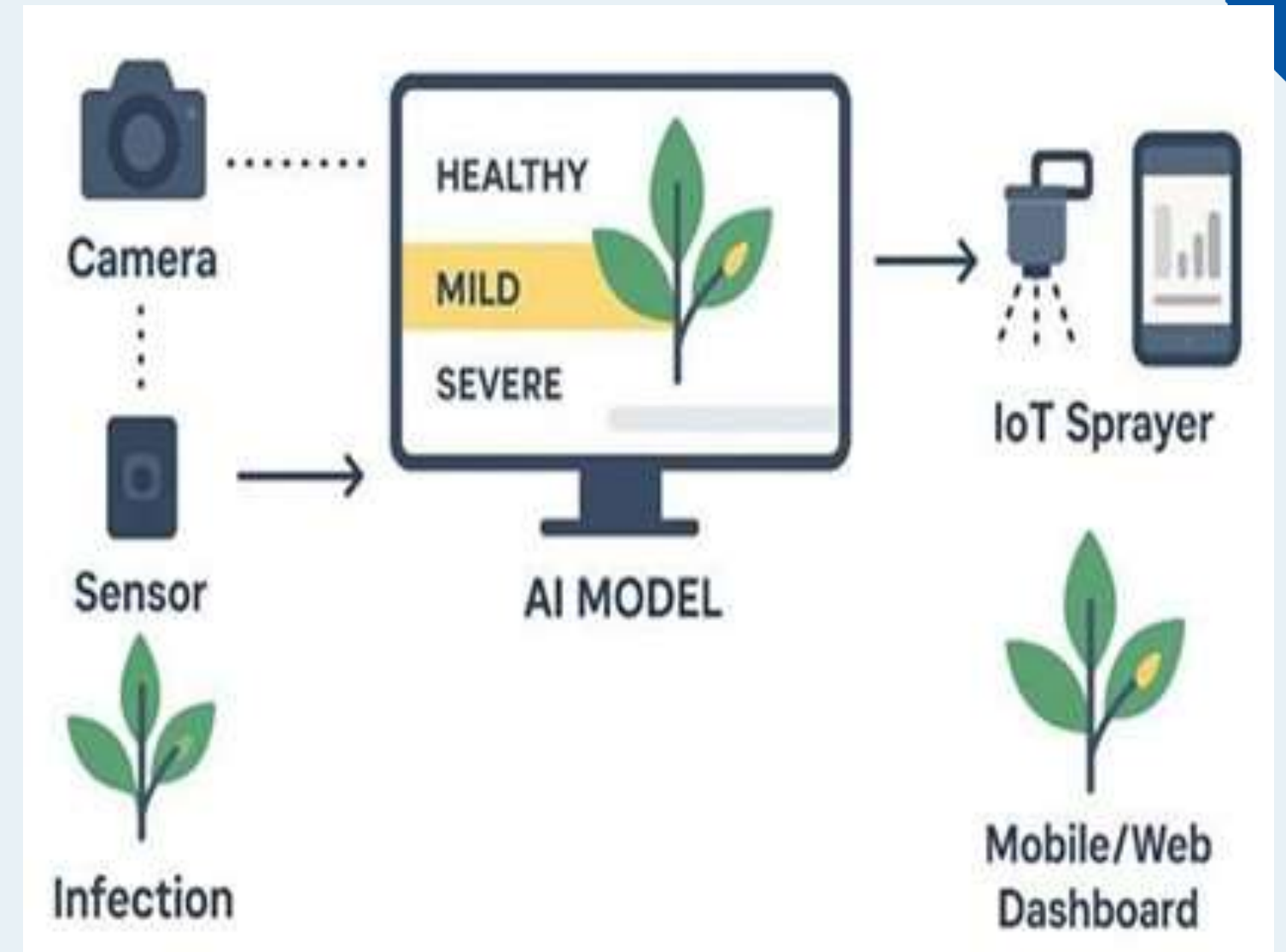
PROPOSED SOLUTION

- Detect plant infection using camera and AI.
- IoT sprayer adjusts dosage according to Low to Medium to High infection.
- Mobile & Web dashboard for farmers to monitor and control remotely



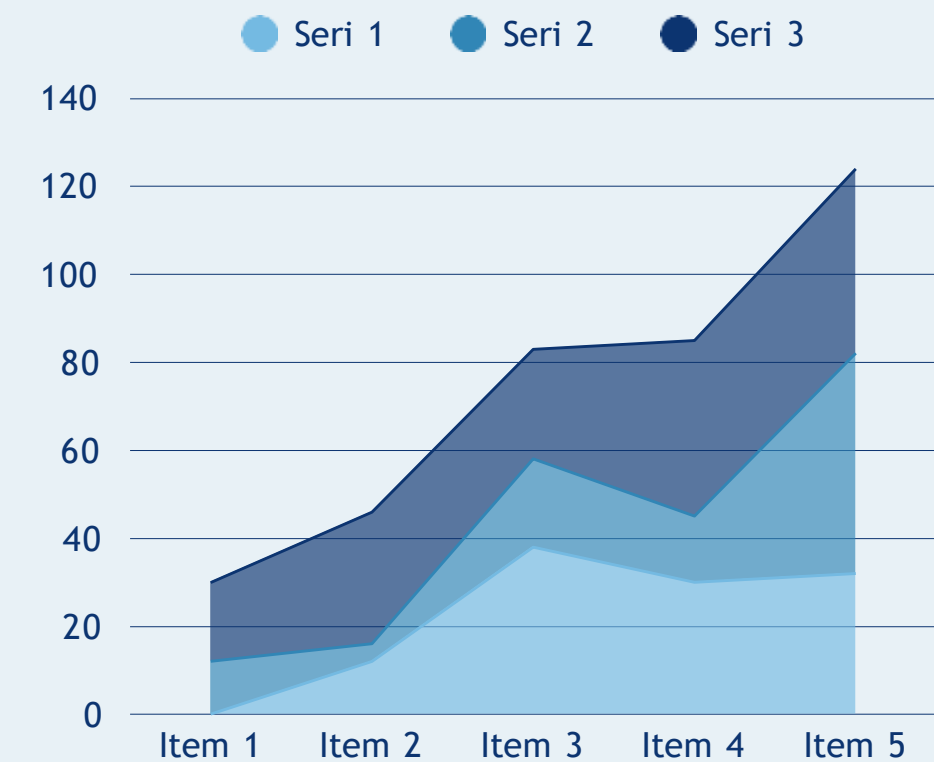
OUR SOLUTION IMPACT:

- Prevents excess spraying, reduces costs and contamination.
- Timely, precise treatment improves yield and quality.



WHAT MAKES US DIFFERENT:

- Beyond hardware unique web platform for easy access anywhere.
- Real-time crop data & history for better decisions



TECHNOLOGIES USED

Hardware

- Camera
- IoT-sprayer
- Environmental sensors
- Arduino microcontroller

Software

- Python for AI/ML, Arduino IDE for hardware.
- TensorFlow & OpenCV for disease detection.
- React & Node.js for mobile/webapp

FLOWCHART

1. Image &
Sensor Data
Collection



2. Train ML Model
(Disease
Detection)



3. Deploy Model
on Edge Device



4. IoT-Controlled
Spray via
App/Dashboard



Farmer
Monitoring
via App/Dashboard

FEASIBILITY

- Technically feasible using AI, IoT and low-cost sensors.
- Affordable prototypes possible for small & large farms.
- Support sustainable agriculture goals.

CHALLENGES

- Accuracy issue
- Hardware damage
- Network issue

IMPACT AND BENEFITS

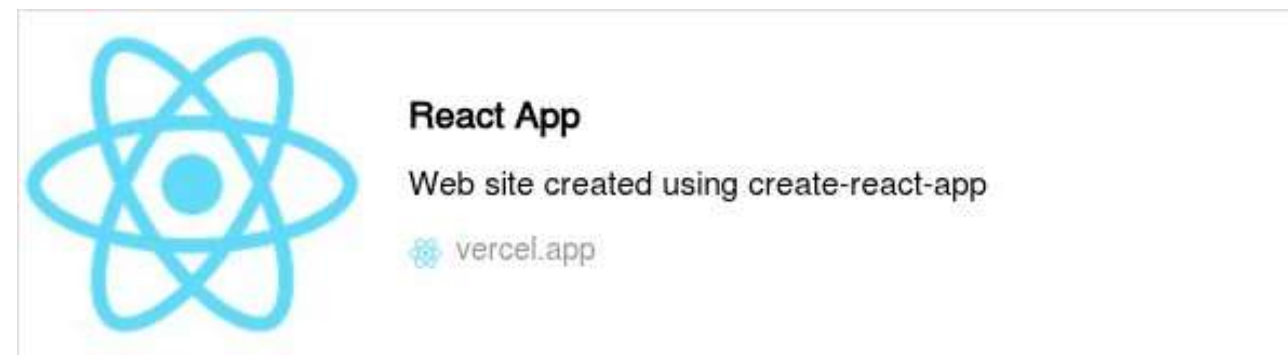
- Reduces pesticide use
- Lower farming costs.
- Protects environment
- Improves crop quality

RESEARCH AND REFERENCE

Research Papers:

- State-of-the-Art Deep Learning Algorithms for Internet of Things-Based Detection of Crop Pests and Diseases:A Comprehensive Review. (10 .1109/ACCESS.2024.3455244)
- Revolutionizing agriculture with artificial intelligence: plant disease detection methods, applications, and their limitations(<https://doi.org/10.3389/fpls.2024.1356260>)

WEBAPP



Farmer Dashboard (WebApp)-[React App](#)



CONCLUSION

- This system uses AI and IoT to spray pesticides efficiently, reduce waste, and support smart agriculture.



THANK YOU